CATERPILLAR®

DEALER BEST PRACTICE SERIES

# Standard Parts Usage Policy for Component Rebuilds

Maintenance and Repair | Application | **Component Renewal (CRC)** | Component Life Management | MARC Management





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## 1.0 Introduction

Every component rebuild requires choices to be made regarding parts consumption during the rebuild process. The choices are basically:

- Reuse part.
- Replace part with a new part.
- Salvage part in house.
- Salvage part using outsourced provider.
- Replace part with Cat Reman.

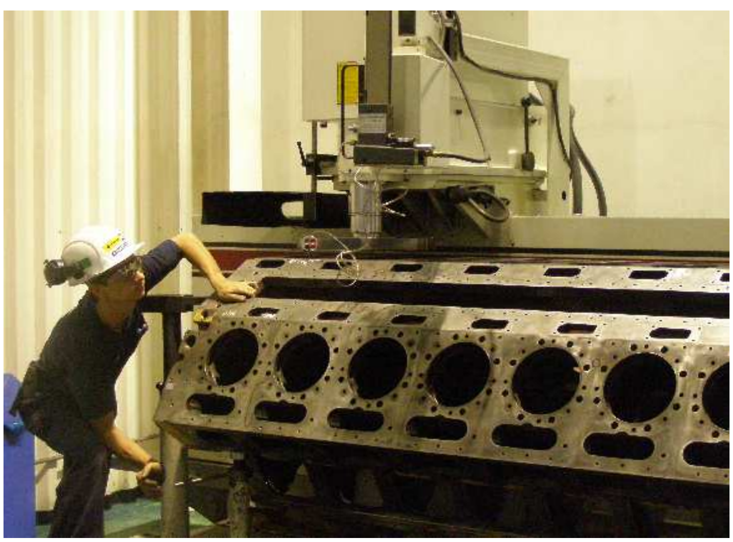
Many times, part usage decisions are made by the technician doing the work. Often these decisions are based on past experience rather than on a formalized company policy and clear guidelines. When technicians make their own decisions without formal CRC policy guidelines, there is often wide variation in parts usage criteria.

Parts consumption is the major cost driver of rebuild components. On average this is equal to 75% or more of overall component rebuild costs.

It is vital to manage parts replacement decisions in a manner which minimizes rebuild costs, standardizes rebuild practices, and delivers the quality and component life expected by the customer.

This Best Practice provides a recommended process to analyze and develop a policy to manage part replacement practices in the CRC based on the cost impact, quality, and turnaround time.

This Best Practice was developed from the conclusions of a worldwide survey conducted with 33 dealers on parts reuse, salvage, and replacement practices.



## 2.0 Best Practice Description

Parts utilized during a major component rebuild will basically fall into three categories:

- Certain parts are always replaced. These typically include gaskets, seals, wear items such as bearings, etc. These parts should become part of a standard rebuild kit. A Best Practice is published on creating standard rebuild kits using the Repair Option Builder program that provides parts lists and replacement percentage for most major components. Defining these kits becomes a CRC "always replace" policy. (See *7.0 Related Best Practices)*
- Certain parts are almost always inspected and reused when meeting reuse criteria. These would include such items as bolts, tubes, housings, etc.
- Certain parts must be either salvaged or replaced.

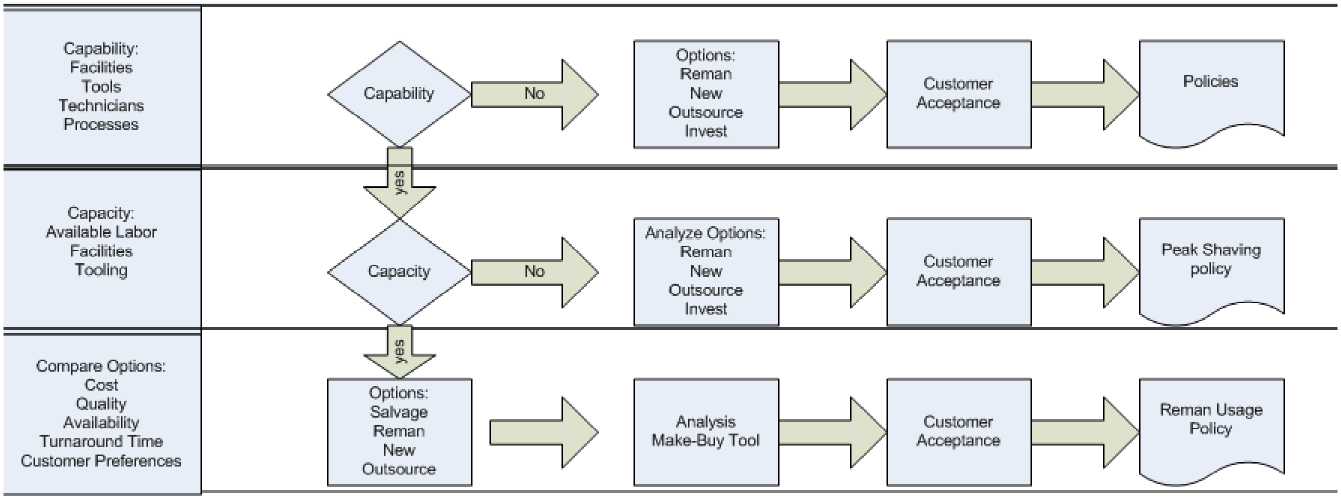
This Best Practice focuses on those parts that require salvage or replacement, analysis of the options, and choosing the best option.

## 3.0 Implementation Steps

The recommended process to analyze and develop a rebuild parts usage policy by major component category follows five basic steps.

- List parts or assemblies that require salvage or replacement during a rebuild. As example, for a 3516 engine the list could include block, crankshaft, cylinder packs, turbos, coolers, manifolds, starter, alternator, water pump, fan drive, etc.
- Does the CRC have the capability to perform the needed salvage? Facility, tooling, processes.
- Does the CRC have the capacity to perform needed salvage? Technicians, facility, resources.
- Compare a cost analysis of the options. Salvage costs versus other options, such as outsource or Reman.
- Determine customer preferences and acceptance. What will the customer accept and be willing to pay for? What rebuild warranty does the customer expect?

**Process Map**




The above process map illustrates the thought process behind creating a parts usage policy.

Capability to salvage is determined by having the right tooling, technicians, and processes in place to perform a required salvage. If a CRC does not have capability to do the work, then the options become:

- Replace with Reman
- Replace with a new part
- Outsource the salvage work
- Invest in developing the capability (long term).

Capacity to salvage is determined by having available technicians, tooling, and facility. If the CRC is already working at full capacity, then the part usage policy becomes a peaking shaving strategy.

Where the CRC has both the capability and the capacity to do salvage work, the decisions move to a “Make vs Buy” economic comparison.

**Cost Analysis of the Options**

If the CRC has both the capability and the capacity to perform salvage on a specific subcomponent or part, then the CRC will need to decide which repair practice alternative makes the most economic sense. Typically if the CRC performs the salvage, a dealer would compare costs in terms of parts, labor, and other miscellaneous expenses to the Reman or new parts option. Other considerations would include the impact on rebuild turnaround time, the component quality and life expected from the chosen option, core credit if Reman is used, warranty, and customer acceptance.

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Below is an example of a cost comparison between Reman, new, and rebuild options. This is an example only. Costs for an actual analysis should be based on local landed costs of parts and fully burdened labor rates.

**Reman** **New** **CRC Rebuild**

793C Truck

ATY Serial Number Prefix

Cyilinder Head Rebuild Comparison

Fully Burden Labor Cost: $60.00

| Qty | Part No. | Desciription | US DN |
|---|---|---|---|
| | Reman | | |
| 1 | 10R7765 | Head Gp - Cylinder | $ 501.88 |
| 1 | 10R7766 | Head Gp. - Cylinder UTN | $ 758.93 |
| **Total Cost - Reman** | | | $ 501.88 |
| **Total Cost - Reman UTN** | | | $ 758.93 |

| Qty | Part No. | Description | US DN |
|---|---|---|---|
| | New | | |
| 1 | 145-3216 | Head Gp - Cylinder | $1,229.99 |
| **Total Cost - New** | | | $1,229.99 |

| Reuse Rate | Qty | Part No. | Description | DN | Ext DN |
|---|---|---|---|---|---|
| | | Rebuild | | | |
| 100% | 2 | 133-9306 | GUIDE-VALVE | $ 8.62 | $ 17.24 |
| 100% | 2 | 194-4897 | VALVE-EXHAUST | $ 41.06 | $ 82.12 |
| 100% | 2 | 197-6995 | GUIDE-VALVE | $ 7.57 | $ 15.14 |
| 100% | 2 | 197-7006 | SEAL | $ 1.52 | $ 3.04 |
| 100% | 2 | 197-7062 | ROTOCOIL AS | $ 6.74 | $ 13.48 |
| 100% | 2 | 197-7063 | ROTOCOIL AS | $ 6.92 | $ 13.84 |
| 100% | 2 | 209-0962 | WASHER | $ 3.34 | $ 6.68 |
| 100% | 2 | 210-2542 | VALVE-INLET | $ 32.33 | $ 64.66 |
| 100% | 4 | 2A-4429 | LOCK-RETAINER | $ 0.31 | $ 1.24 |
| 100% | 4 | 194-4902 | SPRING | $ 4.15 | $ 16.60 |
| 100% | 4 | 197-7055 | LOCK-RETAINER | $ 0.38 | $ 1.52 |
| 100% | 4 | 281-6157 | SPRING | $ 8.35 | $ 33.40 |
| 100% | 1 | 8T-6759 | PLUG-PIPE | $ 1.81 | $ 1.81 |
| **Parts** | | | | | $ 270.77 |
| **Miscellaneous** | | | | | $ - |
| **Labor** | | | | Hours | |
| | Cleaning | | | 1 | $60.00 |
| | Salvage | | | 2 | $120.00 |
| | Disassemble and assemble | | | 1 | $60.00 |
| | | | | 4 | $240.00 |
| **Total Cost - Parts and Labor** | | | | | $ 510.77 |

In the above example, CRC salvage was approximately equal to Reman due to the $60.00 per hour labor cost rate. The dealer could then select an option based on other factors, such as workload, turnaround time, warranty, etc.

Once the options and costs are known and compared, the CRC can develop a preferred usage policy.

Most dealers discuss standard rebuild practice with their customers and come to an agreement on how the component rebuilds are performed. Customers may not always agree with the 'lowest cost" option and may express preference for different options at an additional cost.

Example of customer preferences: (customer names excluded)

**Customer: XXXXXX**

General:

Use of REMAN is accepted in all the cases

Engine Assembly & Disassembly

Aftercooler

For 3300 and 3400, use reman or new.

Oil Cooler

Consider use of REMAN and use NEW in the case that this part does not have credit core

**Customer: XXXXXXXX**

General:

Use of REMAN is accepted in all the cases

Adjustments

For block and crankshafts, they will select the most economic option (new or REMAN with or without credit core). The first approach will be REMAN unless the REMAN cost more than 60% of a NEW one

Camshafts

Replace camshafts in every major repair for 793C,793B,785C D11R. Use REMAN for these replacements

Oil Coolers

Replace always and use REMAN or NEW

At the end of this process, the dealer will be able to set clear shop polices on parts usage that manage rebuild cost and quality in a consistent manner.

The policy then becomes part of the work order and part of the "Standard Rebuild Kits".

Example - **Dealer Policy**

- Standard engine rebuild includes Reman alternator, starters, water pumps, coolers, injectors, and turbo cartridges.
- Rebuild Cylinder Heads first rebuild, Reman on second.
- Rebuild Liner Pack first rebuild, Reman on second.
- Camshafts and crankshafts replaced with Reman if full core credit applies to core.

Example – **Customer Policy**

- Always accept Reman except for injectors - replace with new.
- Always buy new parts if full core credit does not apply.

## 4.0 Benefits

- Parts reuse decisions are based on a consistent policy, not individual preferences.
- Manage rebuild cost.
- Manage rebuild quality and consistency.
- Support customer cost per hour and availability needs.
- Provides CRC policy guidance for technicians and process consistency.

## 5.0 Resources Required

- Management time to analyze / develop / implement a policy.
- The policy needs to be re-evaluated on a periodic basis for possible adjustments when parts prices and labor rates change, when new or improved Reman product is introduced, or when CRC workloads create capacity concerns.

## 6.0 Supporting Attachments / References

N/A

## 7.0 Related Best Practices

*(0607-4.4-1082) Creating Standard Component Rebuild Parts Kits*
*(1006-4.2-1030) Improving CRC Turnaround through Machine Shop Subcomponent Exchange Program*
*(1006-4.5-1028) Managing Parts Reuse/Replace Decisions with Dedicated Resource*

## 8.0 Acknowledgements

This BP was created from the result of a 2008 dealer survey and input from several dealers on parts usage policies.

Photos are from Ferreyros, Lima Peru

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